

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

MARK SCHEME

CLASS

BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

20 September 2018

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper

Section B

Answer any **one** question on the separate Answer Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
Section B	
Total	

This document consists of **14** printed pages.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 Malaria is an infectious disease caused by *Plasmodium*. *Plasmodium* requires two hosts to complete its complex life cycle. One of the hosts is the *Anopheles* mosquito, which acts as a vector of malaria.

Transmission of malaria occurs when females of some species of *Anopheles* take blood meals from humans infected with *Plasmodium*, and then feed on uninfected individuals.

Both male and female *Anopheles* mosquitos have piercing and sucking mouthparts. The female mosquito is shown in **Fig. 1.1**.

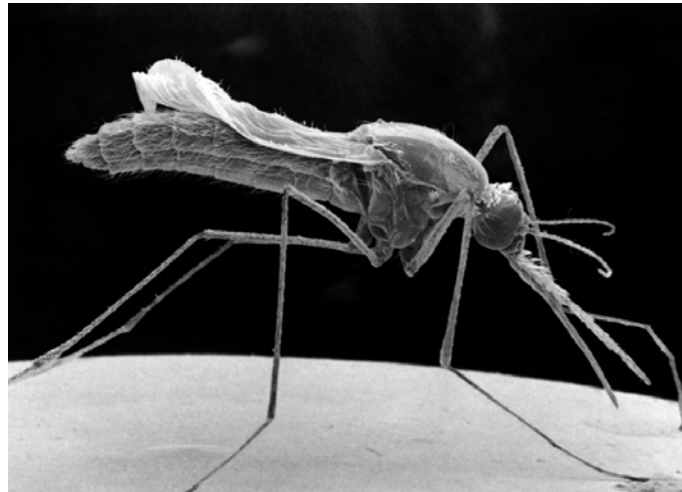


Fig.1.1.

- (a) *Plasmodium* are responsible for millions of infections each year in tropical and subtropical areas of the world. According to the World Health Organization, malaria incidence has increased significantly over the past 50 years, turning this infection into the most important mosquito-borne disease in the world and a global health challenge.

Fig. 1.2 shows the general global distribution of malaria in the year 2005.

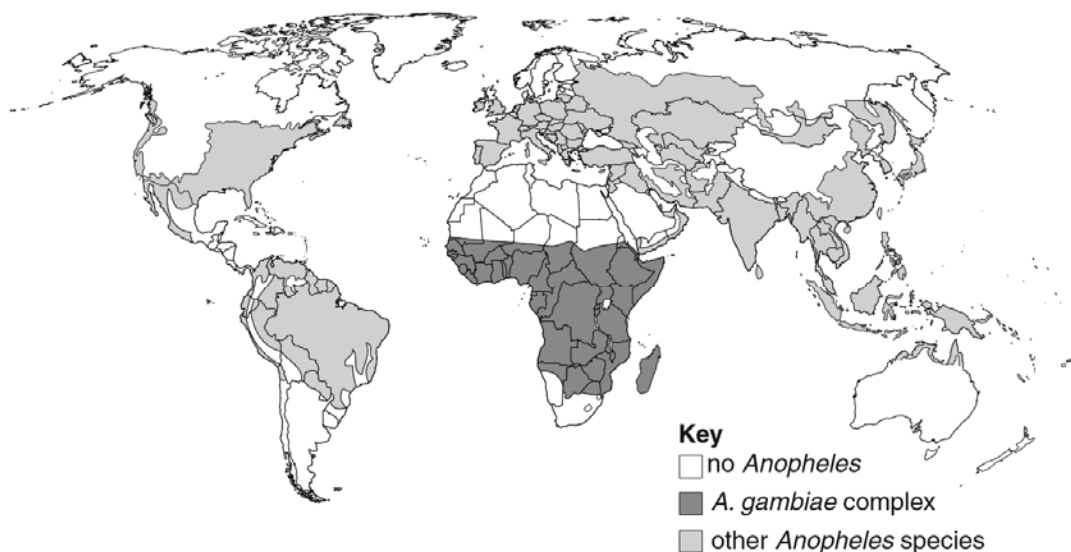


Fig.1.2.

(i) Climate is one of the important factors that affects the distribution of malaria.

Predict and justify the expected distribution of malaria by the end of the 21st century.

[ACJC 2017 P3Q1]

1. Geographical range of Aedes mosquito (vector) and dengue will expand towards the two poles/ to areas infested with mosquitoes;
2. As global temperature is predicted to rise (by 4°C)/ global warming/ climate change due to (increase in greenhouse gas emission);
3. resulting in favourable temperatures for mosquito breeding/ suitable breeding places in the temperate regions/ increased viral load/ higher rate of DENV replication;

[3]

(ii) Symptoms of malaria such as high fever usually develop within 7 to 18 days after being bitten by an infected mosquito and is often associated with muscle pain.

Suggest how the *Plasmodium* may cause these symptoms.

1. Virus stimulates/ activates innate/ non-specific immune response/ immune cells (e.g. dendritic cells) in the first four days;
2. Interferons / cytokines /histamines released result in inflammation, causing pain ;
3. Pyrogen released by activated macrophages leads to rise in systemic body temperature resulting in dengue fever;

Award marks only when reference is made to the respective symptoms

[3]

During an infection, *Plasmodium* will elicit humoral immune response which involves B cell activation as shown in **Fig.1.3**.

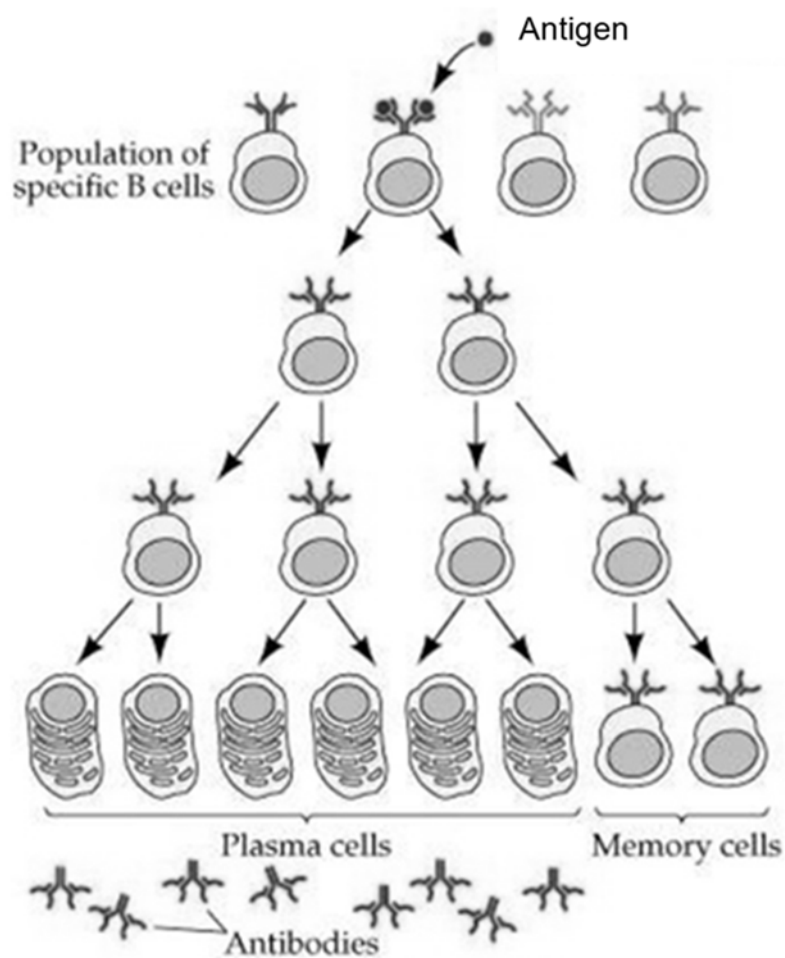


Fig.1.3.

(iii) With reference to **Fig. 1.3** and your own knowledge, explain the significance of mitosis in B cell activation.

Clonal expansion occurs via mitosis to produce genetically identical daughter cells;

This results in increased/ faster production of antibodies/ memory cells/ plasma cells

which can bind to complementary/ specific antigen present

to mediate fast clearance/ destruction of pathogen/ increased rate of phagocytosis/ faster response during secondary infection;

Somatic hypermutation during clonal expansion results in antibodies with increased affinity to antigen;

[3]

- (iv) Memory B cells have long life spans because they do not actively undergo cell division. However, upon activation, a memory B cell can undergo many rounds of proliferation.

Suggest why activated memory B cells can undergo multiple rounds of cell division without dying.

Length of telomeres maintained due to active telomerase/ expression of telomerase upon activation;

® very long length of telomeres

[1]

Fig. 1.3 is part of a complex food web in an area of Kenya where the larvae and adults of *A. gambiae* occur.

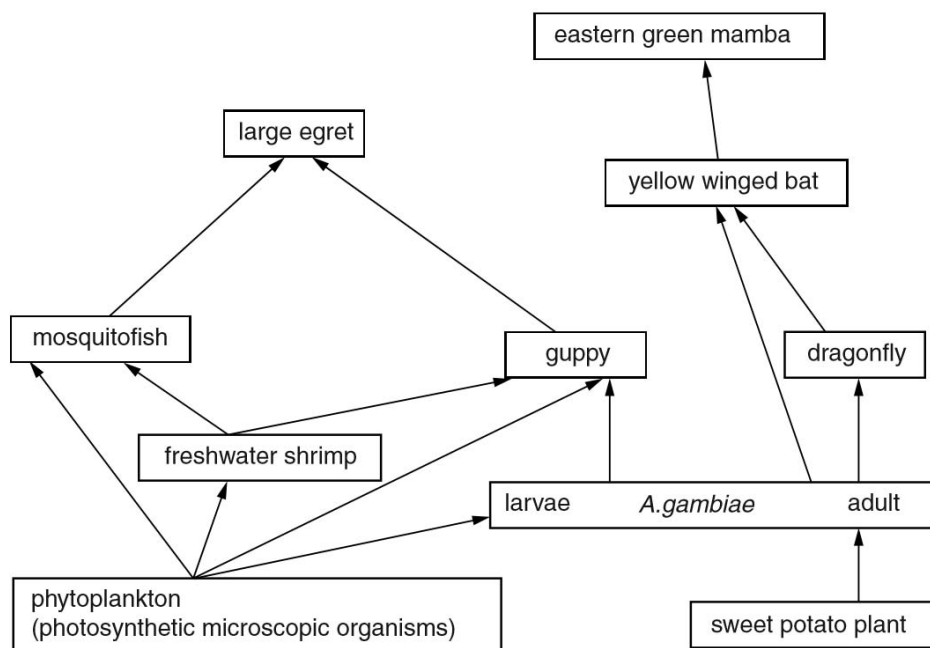


Fig.1.3

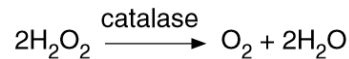
- (v) Suggest how the information in **Fig. 1.3** can be used in the control of malaria in other areas of Kenya.

use biological control ;
 introduce/increase numbers of /AW, predators (of mosquito) / named e.g.
 guppy / dragonfly / yellow winged bats ;
 to eat / consume / reduce number of, mosquitoes ;
 grow crops other than sweet potato / grow less sweet potato ;
A do not grow sweet potato
 numbers (of mosquito) reduce so fewer, act as vectors / feed on humans ;
 AVP ; e.g. use genetic modification to produce sweet potato crop with toxin
 against mosquito

[max 2]

[2]

The enzyme catalase is found in many plant and animal tissues. The enzyme catalyses the decomposition of hydrogen peroxide, which is a toxic product of metabolism. The reaction is:



A research team investigated the activity of two forms of catalase, P and Q, extracted from *Anopheles gambiae*, an important vector of malaria. The team investigated the effect of increasing concentrations of hydrogen peroxide on the activity of these two forms of catalase.

The results are shown in **Fig. 1.4**. [9700 CIE w15 qp 22 Q5]

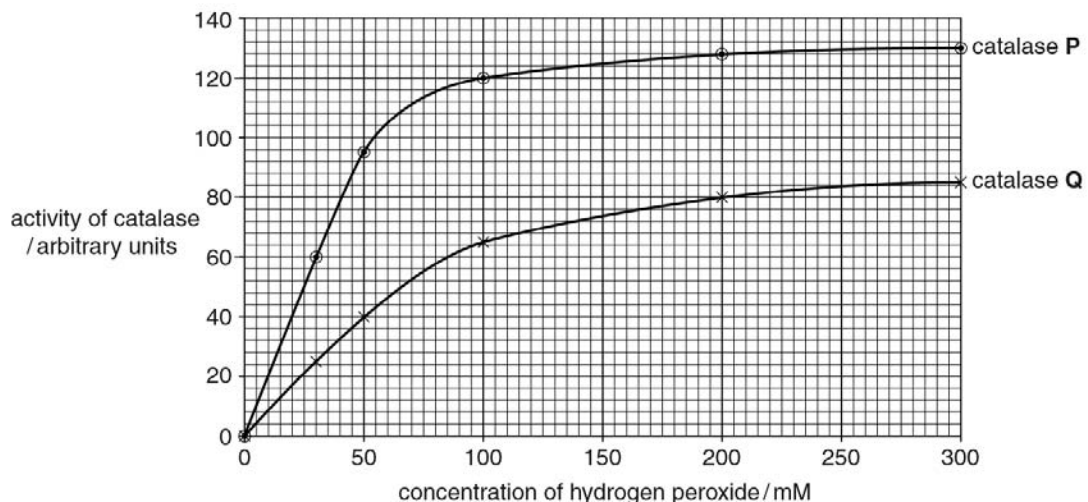


Fig. 1.4

(b) With reference to **Fig. 1.4**, describe and explain the effect of increasing the concentration of hydrogen peroxide on the activity of catalase P.

(a) *description*

- 1 activity / rate, increases to a, maximum / plateau ;
A 'levels off' / remains constant / reaches $V_{\max}$
- 2 increase in, activity / rate, slows ;
- 3 data quote with units to support any correct statement ;
e.g. mp 1128–132 au at 250–300 mM
e.g. mp 20 to 120 au between 0 and 100 mM, 120–128 au between 100 and 200 mM
A au for arbitrary units

explanation

at low / increasing, concentration of hydrogen peroxide

- 4 substrate / hydrogen peroxide, (concentration) is limiting (factor) ;
- 5 active sites, unoccupied (low concentration) / become more occupied (increasing concentration) ;
R active side (*penalise once*)
- 6 (low concentration) few collisions between enzyme and substrate / few ESC formed
or
(increasing concentration) more collisions between enzyme and substrate / increasing ESC formed ;

at high (activity slows) / higher (plateau) concentration of hydrogen peroxide

- 7 enzyme / catalase, concentration / AW, becomes / is, limiting (factor) ;
- 8 maximum number of enzyme-substrate complexes formed ;
A ES complexes / ESCs
- 9 (all) active sites, saturated / (always) occupied ; A ora

[max 5]

- (c) Each molecule of catalase consists of four identical polypeptides. The two forms of catalase in *A. gambiae* differ by only one amino acid at position 2 in the amino acid sequence. Catalase P has serine and catalase Q has tryptophan.

Suggest how the difference in one amino acid is responsible for the lower activity of catalase Q compared with catalase P.

Answer as below:

amino acid at position 2, is part of active site / helps to give shape to active site / helps form the structure of the active site ;

plus one from:

idea of different, R group / side chain, gives different properties ;

A tryptophan has a, hydrophobic / larger, R group / serine has a polar R group, different properties ;

(slightly) different, folding of polypeptide / secondary structure / tertiary structure / active site / catalytic site / binding site ;

suggested reasons e.g. electrons less easily transferred

ref. to induced fit , more efficient with P ; ora

different interactions between polypeptides (in catalase) ;

[2]

- (d) Female mosquitoes feed on blood in order to produce their eggs. After feeding, the metabolic rate increases for egg production.

The researchers allowed female mosquitoes to feed on blood. They found that female mosquitoes with only catalase P produced more eggs than those with only catalase Q.

Suggest why there is a difference in egg production between the two types of *A. gambiae*.

- 1 increased, metabolic rate / protein metabolism (after feeding) means, increased / more, hydrogen peroxide (produced) ;
- 2 *idea that* less effective, catalase / **Q**, means, more hydrogen peroxide remains / less hydrogen peroxide broken down ; **ora**
more hydrogen peroxide from increased metabolism is broken down faster in P = 2 marks
- 3 hydrogen peroxide, interferes with / is damaging to / **AW**, egg production ;
- 4 **AVP** ;
I ref. to oxygen production and use in aerobic respiration

- (e) Copper ions can act as a non-competitive inhibitor of catalase.

Explain how copper ions can be used in the fight against malaria.

Bind to allosteric site / site other than active site ;

Causes change in (shape of) active site ;

@ changes shape in active site (so) substrate cannot bind (to enzyme / active site) / ES complex cannot form ;

If *A. gambiae* is exposed to copper ions, catalase is inhibited ;

Cannot break down hydrogen peroxide, which is a toxic product of metabolism

Kills male mosquitoes then cannot fertilise female mosquitoes so less eggs and population of vectors decreased ;

Poison female mosquitoes then less females to bite humans and decrease spread of malaria ;

[4]

- (f) Malaria is an important disease worldwide. **Table 1.1** shows recent information about malaria cases reported during one year in six different countries.

Table. 1.1

country	region	number of cases	number of cases per 100 000 population
Germany	Europe	4000	5
India	Asia	2 300 000	185
Japan	Asia	27 000	21
South Africa	Africa	490 000	981
Swaziland	Africa	15 000	1287
United Kingdom	Europe	7900	13

With reference to **Table 1.1**, explain the advantage of calculating the number of cases of malaria per 100 000 population rather than stating the number of cases alone.

[9184 CIE s14 qp 23]

(number of cases per 100000) shows, proportion/AW, of population affected ; AW

idea that easier to visualise, the severity of the problem ;

useful / more reliable, qualified ; e.g. for making comparisons between different countries with different populations

(as) countries with larger populations will usually have more cases /higher number of cases may just mean larger population of country ;

comparative data quote to support ;

- (g) Suggest why malaria is more likely to be fatal in people who have HIV/AIDS than in those who do not have HIV/AIDS.

(HIV/AIDS leads to) weak immune system/reduced immunity (to disease) ;

detail ; e.g. reduced action of phagocytes / Th lymphocytes low in number / B-lymphocyte response low

(so) pathogens, can multiply faster/ are not destroyed before they cause disease ;

idea that important, organs / systems, may already be suffering from consequences of HIV/AIDS (so more likely to stop functioning) ;

[2]

[Total: 29]

- 2 In anaerobic respiration in yeast, the pyruvate molecules are broken down to produce ethanol and carbon dioxide. The release of carbon dioxide can be used to investigate the rate of anaerobic respiration. [H1 2015 Q2(b)]

Fig. 2.1 shows an experiment which was set up to find the rate of anaerobic respiration.

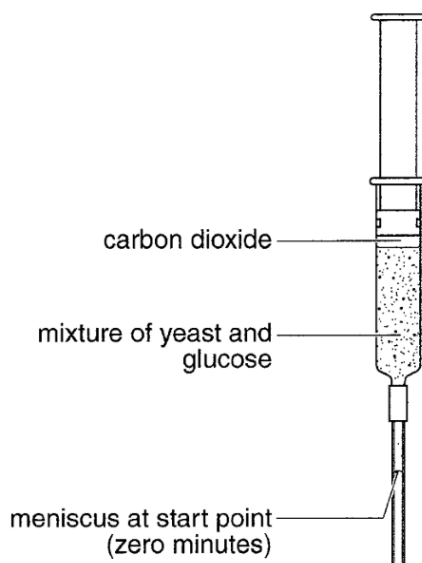


Fig. 2.1

The meniscus moves down the tube as carbon dioxide is released.

Table 2.1 shows the distance moved by the meniscus from the start point. This was recorded every 10 minutes.

Table 2.1

time/minutes	0	10	20	30	40	50	60	70	80	90
distance travelled by meniscus from start point/mm	0	1	2	5	9	14	21	45	73	98

The rate of anaerobic respiration can be calculated by using the rate of movement of the meniscus.

Calculate the rate of anaerobic respiration between 70 and 80 minutes.

You will lose marks if you do not show your working.

$$\frac{73 - 45}{10} = \frac{28}{10} = 2.8$$

2.8 ;

Rate of anaerobic respiration mm min⁻¹ [2]

- (a) The experiment was repeated three more times. Each time, the glucose (a monosaccharide) was replaced with a different disaccharide sugar:
- Maltose – a disaccharide of glucose and glucose.
 - Sucrose – a disaccharide of glucose and fructose.
 - Lactose – a disaccharide of glucose and galactose.

Table 2.2 (a), (b) and (c) show the results of these experiments.

Table 2.2 (a): Using maltose

time/minutes	0	10	20	30	40	50	60	70	80	90
distance travelled by meniscus from start point/mm	0	0	0	0	0	2	3	6	9	12

Table 2.2 (b): Using sucrose

time/minutes	0	10	20	30	40	50	60	70	80	90
distance travelled by meniscus from start point/mm	0	0	0	1	3	11	22	37	48	61

Table 2.2 (c): Using lactose

time/minutes	0	10	20	30	40	50	60	70	80	90
distance travelled by meniscus from start point/mm	0	0	0	0	0	0	0	0	0	0

With reference to the information provided in **Table 2.2** (a), (b) and (c) and your biological knowledge:

- (i) describe the difference in the results for maltose and sucrose, and suggest one explanation for this difference,

For maltose, the meniscus didn't move from time 0 to 40 minutes;

from 50 minutes to 90 minutes, the distance moved increased proportionally from 2m to 12mm ;

For sucrose, the meniscus didn't move from time 0 to 30 minutes / moved much earlier than maltose ;

Moved rapidly increasing from 1mm at time 30 minutes to 61mm at time 90 minutes ;

There is more sucrose than maltase (due to higher gene expression) in yeast / sucrose produces glucose and fructose and fructose is an intermediate of glycolysis and can be metabolised faster ;

Reference: file:///C:/Users/S8103694B/Downloads/184804-Article%20Text-191515-1-10-20140606%20(1).pdf

[3]

(ii) Suggest **two** explanations for the results for lactose.

Yeast lacks lactase ;

Yeast need time more than 90 minutes to express the enzymes required to break down lactose ;

AVP ;

[2]

(b) During anaerobic respiration in mammals, pyruvate is broken down by a different pathway from that in yeast.

Outline what happens to pyruvate in mammals during anaerobic respiration.

Pyruvate is broken down by lactate dehydrogenase;

To produce lactate ;

Oxidation of pyruvate ;

Addition of 2 hydrogen atoms ;

which in turn convert reduced NAD to oxidised NAD ;

Occurs in cytoplasm ;

[3]

[Total: 10]

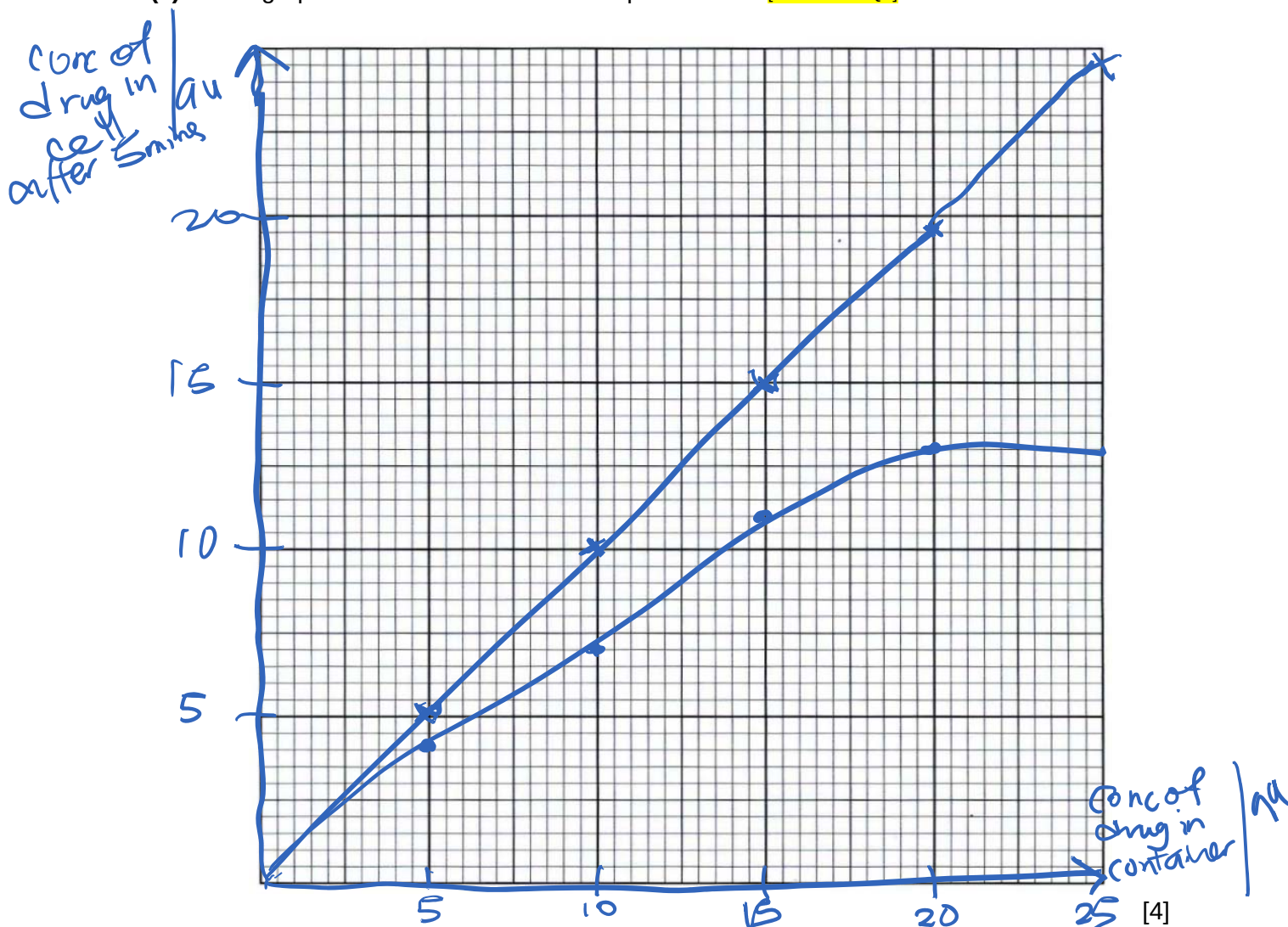
- 3 **Table 3.1** shows the results from an investigation to determine how two different drugs, **A** and **B**, enter animal cells.

Cells which did not contain either drug were placed into separate containers. Different concentrations of each drug were added and after 5 minutes the drug concentration in the cell was measured.

Table 3.1

concentration of drug A in container /arbitrary units	concentration of drug A inside the cells after 5 minutes /arbitrary units	concentration of drug B in container /arbitrary units	concentration of drug B inside the cells after 5 minutes /arbitrary units
0	0	0	0
5	5	5	4
10	10	10	7
15	15	15	11
20	19	20	13
25	24	25	13

- (a) Plot a graph of these data on the same pair of axes. [H1 2011 P2Q4]



- (b) From your knowledge of the cell surface membrane, explain if the information in **Table 3.1** suggests that drug A is lipid soluble.

Yes drug A is lipid soluble ;

With increasing concentration of drug in container, the concentration of drug within the cell after 5 minutes increased as well ;

With 5 a.u. of drug outside the cell, there is 5 a.u. of drug within the cell after 5 minutes ;
[Award mark for making reference to the data in table 3.1.]

The transfer of the drug from container into the cell does not seem to require any process or

[4]

- (c) Suggest, with reasons, how drug B is transported into the cells.

Drug B is transported into the cell via facilitated diffusion ;

The use of transport proteins

Drug B is charged / hydrophilic ; cannot pass through the hydrophobic core of the membrane bilayer ;

When there is 20 a.u. of drugs outside the cells, there is 13 a.u. of drug into the cells after 5 minutes ;

When the concentration of drugs outside the cells is increased to 25 a.u., there is still 13 a.u. of drugs inside the cells after 5 minutes ;

This implies that the drug was being transported across the cell membrane via transport proteins ;

Saturation of transport proteins hence concentration of drug remains at 13 a.u.

Examiners' comments: Candidates found this question more challenging. Some candidates used the graph to help them answer the question, clearly using their knowledge and understanding of facilitated diffusion. Other candidates did not appreciate that this could not be active transport as at no time was the drug entering against a concentration gradient.

[3]

[Total: 11]

Section B

Answer **one** question in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4(a)** The cell theory is based on a number of principles. [9]
 Discuss the extent, in humans, to which these principles can be accounted for by mitosis and the different types of stem cells.
 You should consider both individual humans and populations of humans from one generation to the next.
- (b)** Outline the roles of hydrogen bonds in biomolecules. [6]
- (c)** Discuss the suggestion that all living organisms on Earth depend on phosphate. [10]

[Total: 25]

- 5(a)** In 2014, Dalrymple, Buswell and Moles published an articles on their studies of plants and their proteins, after being introduced to New Zealand and Australia. [6]
 They concluded that asexually reproducing species change as rapidly as sexually reproducing species when introduced to a new environment. This conclusion is unexpected. [H1 2017 P2Q6c]
 Explain why this conclusion is unexpected.
- (b)** During the study of proteins, explain how it is possible for both plant amylase and bacterial amylase to break down the same substrate, but for the bacterial amylase to have a higher optimum temperature. [H1 2017 P2Q5c] [6]
- (c)** A comparison of DNA sequences of many genes from a variety of species has shown that cytochrome c is an example of molecular homology. [13]
 Gel electrophoresis can be used to compare DNA that codes for cytochrome c.
 Describe and explain how gel electrophoresis can be used to compare DNA that codes for cytochrome c to show the evolutionary relationships between a variety of species. [H1 2015 P2Q5b]

[Total: 25]

Question	Answer	Marks
7(b)	any eight from marking points 1–15: <i>cell theory</i> 1 living organisms are made up of cells ; 2 all cells come from pre-existing cells ; <i>within individual (max 5):</i> 3 in humans, all body cells are produced by mitosis ; 4 and arise from stem cells ; 5 stem cells themselves arise from stem cells ; 6 description of totipotent > pluripotent > multipotent stem cells in terms of contribution to cell theory ; 7 totipotent stem cells can give rise to all cell types ; 8 zygotic stem cells are totipotent ; 9 pluripotent stem cells give rise to nearly all cell types ; 10 embryonic stem cells are pluripotent ; 11 multipotent stem cells give rise to a limited range of cell types ; 12 blood stem cells are multipotent ; <i>from one generation to the next:</i> 13 gametes are produced by meiosis ; 14 gametes fuse during fertilisation to produce zygote ; <i>comment on extent:</i> 15 stem cells and mitosis account for cell theory within organisms but not from one generation to the next ; QWC: draws a conclusion consistent with the candidate's discussion, with reference to marking points from all three sections for marking points 1–14 ;	9

Question	Answer	Marks
6(a)	any six from: 1 important feature of, secondary/tertiary/quaternary, structure, of proteins ; 2 (helps) holding protein in specific 3-D shape ; 3 allowing specific function ; 4 join together two strands of nucleic acid ; 5 detail, either two DNA strands or strands of DNA and mRNA ; 6 in cellulose ; 7 H bonds between or within chains, provide strength ; 8 AVP ; further detail 9 AVP ; further detail or ref. to water or other examples	6

Question	Answer	Marks
6(b)	any eight from marking points 1–13; cell membranes: 1 all living organisms have cell membranes ; 2 (cell membranes) require phosphate for phospholipids ; 3 help to regulate what, enters/leaves, cells/organelles ; 4 contribute to fluidity of (cell) membrane ; e.g. phagocytosis. etc. nucleic acids: 5 genetic code relies on nucleic acids ; 6 the backbone of nucleic acids is a sugar-phosphate chain ; 7 therefore, without nucleic acids no storage of information ; 8 AVP ; further detail ATP: 9 most energy stored in the bonds of ATP ; 10 ATP more energy rich than ADP ; 11 AVP ; further detail others: 12 AVP ; e.g. role in NADP 13 AVP ; e.g. role in bones QWC: draws an appropriate conclusion with reference to marking points from at least two different sections ;	9
	Answer	Marks

Q6(a) : Examiners' comments: Many candidates considered the significance of introducing species to new environments in terms of novel selection pressures and natural selection. Fewer explained that underpinning these processes was the existence of genetic variation, and the contrasting roles of asexual reproduction and sexual reproduction in generating this variation. A minority of candidates provided well-presented explanations

Q6(b) : Examiners' comments: Many candidates described the different levels of protein structure and used this knowledge to explain the specificity of the mode of action of enzymes. Few went on to consider how differences in structure between human and bacterial amylase could account for differences in their optimum temperatures.

Q6(c) : Examiners' comments: The majority of candidates were able to describe the process of gel electrophoresis. A significant minority did not make clear how the DNA from different species would be compared to show the evolutionary relationships between them